

CSC 212: Data Structures and Abstractions

03: Introduction to Analysis of Algorithms (part 2)

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Fall 2026



From last lecture ...

- Empirical analysis worked
 - ✓ ... and that's exactly the problem
 - ✓ we **measured** two programs and found **poly1**'s time roughly **quadruples** when n doubles, while **poly2**'s roughly **doubles**
 - ✓ to learn that, we had to write both programs, compile them, choose inputs, run them hundreds of times, and trust the machine
 - ✓ ... and the answer was valid only for that compiler, that CPU, that day
- Today
 - ✓ get the same answer without running anything

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Q3-1

- You time an algorithm and record:

n	time (s)
1,000	0.004
2,000	0.016
4,000	0.064
8,000	0.256

- What does the doubling ratio tell you?
 - It grows linearly, $T(n) \approx cn$
 - It grows quadratically, $T(n) \approx cn^2$
 - It grows cubically, $T(n) \approx cn^3$
 - It grows logarithmically, $T(n) \approx c \log n$
 - None of the above

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Answer

- B) It grows quadratically, $T(n) \approx cn^2$

- ✓ each time n doubles, the time is multiplied by 4. If $T(n) = cn^b$, then:

$$\frac{T(2n)}{T(n)} = \frac{c(2n)^b}{cn^b} = 2^b = 4$$

apply log to both sides, so $b=2$

- ✓ can fit the constant from the first row:

$$0.004 = c \cdot 1000^2 \Rightarrow c = 4 \times 10^{-9}$$

- ✓ giving:

$$T(n) \approx 4 \times 10^{-9} n^2 \text{ seconds}$$

Prediction:
 $n = 16,000 \Rightarrow T \approx 1.02s$

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Theoretical analysis

“mathematical models for analyzing time and space complexity”

Why the need for a model

- To count operations, we must agree on the operation cost
- However, HW is hostile to analysis
 - ✓ an integer addition and a cache-missing array read may differ by $\sim 100\times$
 - ✓ same instruction takes different times depending on branch prediction
- We analyze against an idealized machine
 - ✓ the model is a simplification

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The RAM model (random access machine)

- Assumptions
 - ✓ memory is an unbounded sequence of cells
 - ✓ any cell is read or written in 1 time unit, regardless of address
 - ✓ each primitive operation takes 1 time unit
 - ✓ instructions execute sequentially, no parallelism, no pipelining, no cache hierarchy
 - ✓ memory is unbounded, dynamic allocation is free
- Advantages
 - ✓ HW/SW independent $\Rightarrow$ the result is still true on different computers
 - ✓ can reason about an algorithm before implementing it
 - ✓ constant factors are hidden, but growth rates survive

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Primitive operations

- A primitive operation is a low-level instruction with constant execution time
 - ✓ assigning a value to a variable
 - ✓ calling a function
 - ✓ performing an arithmetic operation
 - ✓ comparing two numbers
 - ✓ indexing into an array
 - ✓ following an object reference (dereferencing a pointer)
 - ✓ returning from a function
- Instead of timing, **count** how many primitive operations execute as a function of input size n . Call it $T(n)$, the computational cost.
- What is n ?
 - ✓ number of array elements, number of nodes, number of digits, number of vertices and edges

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Worked example

- Summing an array

```
int sum(int *A, int n) {  
    int sum = 0;  
    for (int i = 0 ; i < n ; i++) {  
        sum = sum + A[i];  
    }  
    return sum;  
}
```

Operation	Count	Comments
assignment	$n + 2$	2 outside the loop, n inside
comparison (less than)	$n + 1$	includes the final failing test
addition	n	
array indexing	n	$A[i]$
increment	n	$i++$
return	1	

$$T(n) = (n + 2) + (n + 1) + n + n + n + 1 = 5n + 4$$

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Constant factors

- Is $T(n) = 5n + 4$ the right answer?

- what if we treat $i++$ as read, add, write (3 ops instead of 1)?
 - $T(n) = 7n + 4$
- neither is wrong, **constant factors** depend on what you chose to count
 - different textbooks count differently, different compilers emit different instruction counts
 - constants are not properties of the algorithm

- What is a property of the algorithm?

- $T(n)$ grows **linearly** (double the input n , double the cost)

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Q3-2

- Two students count operations for the same `sum` function. One gets $T(n) = 5n + 5$, the other $T(n) = 3n + 2$. Who is right?

- A. The first, array indexing must be counted separately
- B. The second, index arithmetic is free on modern CPUs
- C. Both, provided each was internally consistent about what they charged for
- D. Neither, the true answer is $4n + 3$
- E. None of the above, unanswerable without knowing the compiler

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Answer

- C) Both, provided each was internally consistent about what they charged for

- the **coefficient** is an artifact of the accounting convention. The **growth rate** is not. Both students correctly concluded that `sum` has $T(n)$ linear in n
 - we will stop writing $5n + 5$ and start writing $\Theta(n)$

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Counting rules

- Consecutive statements
 - ✓ just add, $T = T_1 + T_2$
- A single loop
 - ✓ sum the body's cost over all iterations, $T = \sum_k (\text{cost of iteration } k)$
 - ✓ if every iteration costs the same, $T = \text{number of iterations} \times \text{cost per iteration}$
- Nested loops: work innermost outward
 - ✓ **independent bounds**: multiply
 - ✓ **dependent bounds**: (inner limit depends on outer index) you need a summation, not a product

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Q3-3

- How many multiplications are performed?

```
for (int i = 0 ; i < n ; i++) {  
    sum = sum * A[i];  
}
```

- A. $n - 1$
- B. n
- C. $n + 1$
- D. n^2
- E. None of the above

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Answer

- B) n
- ✓ the loop variable runs $i = 0, 1, \dots, n - 1$, so the body executes exactly n times, with one multiplication each

$$T(n) = \sum_{i=0}^{n-1} 1 = n$$

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Q3-4

- How many divisions are performed?

```
for (int i = 0 ; i < n ; i++) {  
    for (int j = 0 ; j < n ; j++) {  
        sum = sum / A[j];  
    }  
}
```

- A. $2n$
- B. n^3
- C. $n^2 + n$
- D. n^2
- E. None of the above

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Answer

• D) n^2

- ✓ the inner loop runs n times for each of the n outer iterations. The bounds are **independent** (the inner limit does not mention i), so we multiply

$$T(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} 1 = \sum_{i=0}^{n-1} n = n^2$$

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Q3-5

• How many additions are performed?

```
for (int i = 0 ; i < n ; i++) {  
    for (int j = 0 ; j < n ; j++) {  
        for (int k = 0 ; k < n ; k++) {  
            sum = sum + j;  
        }  
    }  
}
```

- A. $3n$
- B. n^2
- C. n^3
- D. $n^3 + n^2 + n$
- E. None of the above

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Answer

• C) n^3

- ✓ all three bounds are **independent**, so the counts multiply

$$T(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} 1 = n^3$$

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Q3-6

• How many additions are performed?

```
for (int i = 0 ; i < n ; i++) {  
    for (int j = 0 ; j < n*n ; j++) {  
        sum = sum * A[j % n];  
    }  
}
```

- A. n^2
- B. n^4
- C. $2n^2$
- D. n^3
- E. None of the above

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Answer

• D) n^3

✓ the inner bound is n^2 , but it still doesn't depend on i , so the bounds remain **independent** and we still multiply

$$T(n) = \sum_{i=0}^{n-1} \sum_{j=0}^{n^2-1} 1 = n \cdot n^2 = n^3$$

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Q3-7

• How many multiplications are performed?

```
for (int i = 0 ; i < n ; i++) {  
    for (int j = 0 ; j < i ; j++) {  
        sum = sum * A[j];  
    }  
}
```

- A. n^2
- B. $\frac{n(n+1)}{2}$
- C. $\frac{n(n-1)}{2}$
- D. $n^2 - n + 1$
- E. None of the above

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Answer

• C) $\frac{n(n-1)}{2}$

✓ the inner loop runs exactly i times, ranging over $0, 1, \dots, n-1$:

$$T(n) = \sum_{i=0}^{n-1} i = 0 + 1 + 2 + \dots + (n-1) = \frac{n(n-1)}{2}$$

Sanity check: $n = 4 \Rightarrow 0 + 1 + 2 + 3 = 6$ and $\frac{4 \cdot 3}{2} = 6$

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Useful series

$$\sum_{i=1}^n 1 = n$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^{n-1} i = \frac{(n-1)n}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$$

$$\sum_{i=0}^n c^i = \frac{c^{n+1} - 1}{c - 1}, c \neq 1$$

Reference sheet:

<https://tug.org/texshowcase/cheat.pdf>

Times you can halve n
before reaching 1: $\lfloor \log_2 n \rfloor$

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A closed-form for $1 + 2 + \dots + n$

• Let $S = 1 + 2 + 3 + \dots + n$

• Write the same sum in reverse order

• Add the two sequences term by term

✓ verify $2S = n(n+1)$

✓ divide both sides by 2, so:

$$S = \sum_{i=1}^n i$$

$$S = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Sanity check: $n = 4 \Rightarrow 1 + 2 + 3 + 4 = 10$ and $\frac{4 \cdot 5}{2} = 10$

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Q3-8

• A stack of cubes grows layer by layer. Layer k (counting from the top) is a $k \times k$ square of cubes

✓ how many cubes in total, as a function of n (number of layers)?

A. $\frac{n(n+1)}{2}$

B. $\frac{n(n+1)(2n+1)}{6}$

C. n^3

D. $\frac{n^2(n+1)^2}{4}$

E. None of the above

$$\begin{aligned} \sum_{i=1}^n i &= \frac{n(n+1)}{2} \\ \sum_{i=1}^{n-1} i &= \frac{(n-1)n}{2} \\ \sum_{i=1}^n i^2 &= \frac{n(n+1)(2n+1)}{6} \\ \sum_{i=1}^n i^3 &= \frac{n^2(n+1)^2}{4} \\ \sum_{i=0}^n c^i &= \frac{c^{n+1}-1}{c-1}, c \neq 1 \end{aligned}$$

n	number of cubes
1	1
2	5
3	14
4	30
...	...
n	?

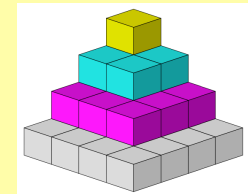


Image credit: Stanford's CS 106B lectures

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Answer

• B) $\frac{n(n+1)(2n+1)}{6}$

✓ Layer k contributes k^2 cubes, so the total is a **sum of squares**:

$$T(n) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\begin{aligned} \sum_{i=1}^n i &= \frac{n(n+1)}{2} \\ \sum_{i=1}^{n-1} i &= \frac{(n-1)n}{2} \\ \sum_{i=1}^n i^2 &= \frac{n(n+1)(2n+1)}{6} \\ \sum_{i=1}^n i^3 &= \frac{n^2(n+1)^2}{4} \\ \sum_{i=0}^n c^i &= \frac{c^{n+1}-1}{c-1}, c \neq 1 \end{aligned}$$

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Case analysis

One input size, many running times

- Some algorithms don't have a single $T(n)$

```
// linear search: return index of key in A, or -1
int find(int *A, int n, int key) {
    for (int i = 0; i < n; i++) {
        if (A[i] == key) return i;    // early exit
    }
    return -1;
}
```

- With $n = 1,000,000$:
 - same algorithm, different cost
 - one function $T(n)$ is not enough

input	comparisons
key at position 0	1
key at position 1,000	1001
key at position 999,999	1,000,000
key absent	1,000,000

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Best, worst, and average case

- Best case
 - cost on the **easiest input of size n**
 - rarely useful alone: every algorithm looks good on its best case
 - its real use: proving an algorithm is **bad**, if even the best case is slow, the algorithm is slow
- Worst case
 - cost on the **hardest input of size n**
 - never worse, on any input, essential wherever there's a deadline
- Average case
 - expected cost over a probability distribution of inputs**
 - requires to **state the distribution**, and real inputs are rarely uniform
 - mathematically the hardest of the three

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Worked example

- Assumptions
 - key is present (successful search)
 - key is equally likely to be at any of the n positions (uniform)

```
int find(int *A, int n, int key) {
    for (int i = 0; i < n; i++) {
        if (A[i] == key) return i;
    }
    return -1;
}
```

If the key sits at position i (0-indexed), the loop performs $i + 1$ comparisons
Each position has probability $\frac{1}{n}$

$$\mathbb{E}[T] = \sum_{i=0}^{n-1} \frac{1}{n} (i + 1) = \frac{1}{n} \sum_{k=1}^n k = \frac{1}{n} \frac{n(n+1)}{2} = \frac{n+1}{2}$$

Best: 1 comparison, Average: $\frac{n(n+1)}{2}$, Worst: n

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Q3-9

- For linear search on an array of n distinct elements, which statement is False?
 - A. The best case is 1 comparison
 - B. The worst case is n comparisons
 - C. The average case for a successful uniform search is $\frac{n+1}{2}$
 - D. The average case always lies exactly halfway between the best and worst cases

Image credit: Stanford's CS 106B lectures

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Answer

- D) is false

✓ $\frac{n+1}{2}$ happens to sit near the midpoint, but that is a coincidence of this algorithm and this distribution

- ✓ **Counterexample:** suppose the key is present with probability only 0.01. Then:

$$\mathbb{E}[T] = 0.99n + 0.01 \cdot \frac{n+1}{2} \approx 0.995n$$

- which is essentially the **worst** case, nowhere near the midpoint

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Q3-10

- Fill in the table. Count the operation that dominates each algorithm

Algorithm	Best Case	Average Case	Worst Case
Find value in unsorted sequence			
Find largest in unsorted sequence			
Find largest in sorted sequence			
Factorial (iterative)			

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Answer

Three of the four rows have identical best, average and worst cases

Algorithm	Best Case	Average Case	Worst Case
Find value in unsorted sequence	$T(n) = 1$	$T(n) = \frac{n+1}{2}$	$T(n) = n$
Find largest in unsorted sequence	$T(n) = n$	$T(n) = n$	$T(n) = n$
Find largest in sorted sequence	$T(n) = 1$	$T(n) = 1$	$T(n) = 1$
Factorial (iterative)	$T(n) = n$	$T(n) = n$	$T(n) = n$

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Space complexity

- Time is not the only resource
- Auxiliary space
 - ✓ memory used beyond the input itself
 - ✓ “total” and “auxiliary” differ by exactly the input size
- Why it matters?
 - ✓ an in-place sort handles a 10 GB file; one that copies does not
 - ✓ embedded and mobile targets have hard memory ceilings
 - ✓ there is almost always a time–space tradeoff

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Q3-11

- Which uses $\Theta(1)$ auxiliary space?

```
// (I)
int max1(vector<int>& A) {
    int m = A[0];
    for (int x : A) if (x > m) m = x;
    return m;
}

// (II)
int max2(vector<int>& A) {
    vector<int> B = A;
    sort(B.begin(), B.end());
    return B.back();
}
```

- A. I only
- B. II only
- C. Both
- D. Neither
- E. None of the above

Answer

- A) I only
 - ✓ **max1** holds one extra `int` regardless of $n \rightarrow \Theta(1)$ auxiliary space, $\Theta(n)$ time
 - ✓ **max2** copies the entire array $\rightarrow \Theta(n)$ auxiliary space, $\Theta(n \log n)$ time
 - it is correct, it passes every test, it does more work than necessary